

AIDS-1 Module 3

What is Unification? Give example. (5 marks)

Unification is the process of finding a substitution that makes two logical expressions identical. It is a fundamental operation in automated theorem proving and logic programming.

Key points:

- A substitution maps variables to terms
- Two expressions are unifiable if there exists a substitution that makes them syntactically equal
- The most general unifier (MGU) is the simplest such substitution

Example:

Unify the expressions: $P(x, f(y))$ and $P(a, f(z))$

The substitution $\{x/a, z/y\}$ makes both expressions become $P(a, f(y))$, so they are unified.

Another example: $\text{Loves}(x, y)$ and $\text{Loves}(\text{John}, \text{Mary})$ unify with $\{x/\text{John}, y/\text{Mary}\}$.

What is Skolemization? Explain Skolem constant and Skolem function. (5 marks)

Skolemization is the process of removing existential quantifiers from a first-order logic formula by replacing existentially quantified variables with Skolem constants or Skolem functions.

Skolem constant: Used when an existential quantifier is not within the scope of any universal quantifier. The existential variable is replaced by a new constant symbol not appearing elsewhere.

Example: $\exists x P(x)$ becomes $P(c)$ where c is a Skolem constant.

Skolem function: Used when an existential quantifier appears inside the scope of universal quantifiers. The existential variable becomes a function of the universally quantified variables.

Example: $\forall y \exists x P(x, y)$ becomes $\forall y P(f(y), y)$ where f is a Skolem function.

The law says it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove that Col. West is a criminal! (10 marks)

Step 1: Represent the knowledge in First-Order Logic

1. $\forall x \forall y \forall z (American(x) \wedge Weapon(y) \wedge Sells(x, y, z) \wedge Hostile(z) \rightarrow Criminal(x))$
2. $\exists x Missile(x) \wedge Owns(Nono, x)$
3. $\forall x (Missile(x) \wedge Owns(Nono, x) \rightarrow Sells(West, x, Nono))$
4. $\forall x (Missile(x) \rightarrow Weapon(x))$
5. $\forall x (Enemy(x, America) \rightarrow Hostile(x))$
6. $American(West)$
7. $Enemy(Nono, America)$

Step 2: Convert to CNF

- From 2: $Missile(M1), Owns(Nono, M1)$
- From 3: $\neg Missile(x) \vee \neg Owns(Nono, x) \vee Sells(West, x, Nono)$
- From 4: $\neg Missile(x) \vee Weapon(x)$
- From 5: $\neg Enemy(x, America) \vee Hostile(x)$
- From 6: $American(West)$
- From 7: $Enemy(Nono, America)$
- Negation of conclusion: $\neg Criminal(West)$

Step 3: Resolution proof

1. $\neg American(x) \vee \neg Weapon(y) \vee \neg Sells(x, y, z) \vee \neg Hostile(z) \vee Criminal(x)$
(From 1)
2. $\neg Missile(x) \vee Owns(Nono, x)$ is not used (correction: $Missile(M1), Owns(Nono, M1)$ are facts)
3. $\neg Missile(x) \vee \neg Owns(Nono, x) \vee Sells(West, x, Nono)$
4. $\neg Missile(x) \vee Weapon(x)$
5. $\neg Enemy(x, America) \vee Hostile(x)$
6. $American(West)$
7. $Enemy(Nono, America)$
8. $Missile(M1)$
9. $Owns(Nono, M1)$
10. $\neg Criminal(West)$ (Negated goal)

Resolve 3 with 8: $\neg \text{Owns}(\text{Nono}, \text{M1}) \vee \text{Sells}(\text{West}, \text{M1}, \text{Nono})$ (using $x = \text{M1}$)

Resolve with 9: $\text{Sells}(\text{West}, \text{M1}, \text{Nono})$

Resolve 4 with 8: $\text{Weapon}(\text{M1})$

Resolve 5 with 7: $\text{Hostile}(\text{Nono})$

Now resolve 1 with 6: $\neg \text{Weapon}(y) \vee \neg \text{Sells}(\text{West}, y, z) \vee \neg \text{Hostile}(z) \vee \text{Criminal}(\text{West})$ ($x = \text{West}$)

Resolve with $\text{Weapon}(\text{M1})$: $\neg \text{Sells}(\text{West}, \text{M1}, z) \vee \neg \text{Hostile}(z) \vee \text{Criminal}(\text{West})$

Resolve with $\text{Sells}(\text{West}, \text{M1}, \text{Nono})$: $\neg \text{Hostile}(\text{Nono}) \vee \text{Criminal}(\text{West})$

Resolve with $\text{Hostile}(\text{Nono})$: $\text{Criminal}(\text{West})$

Resolve with 10: Empty clause (contradiction)

Therefore, Colonel West is a criminal.

Marcus problem using resolution (10 marks)

Given facts:

1. $\text{Man}(\text{Marcus})$
2. $\text{Pompeian}(\text{Marcus})$
3. $\forall x (\text{Pompeian}(x) \rightarrow \text{Roman}(x))$
4. $\text{Ruler}(\text{Caesar})$
5. $\forall x (\text{Pompeian}(x) \rightarrow \text{LoyalTo}(x, \text{Caesar}) \vee \text{Hates}(x, \text{Caesar}))$
6. $\forall x \exists y \text{LoyalTo}(x, y)$
7. $\forall x \forall y (\text{Person}(x) \wedge \text{Ruler}(y) \wedge \text{TryAssassinate}(x, y) \rightarrow \neg \text{LoyalTo}(x, y))$
8. $\text{TryAssassinate}(\text{Marcus}, \text{Caesar})$

Question: Was Marcus loyal to Caesar? (Prove or disprove $\text{LoyalTo}(\text{Marcus}, \text{Caesar})$)

Convert to CNF:

1. $\text{Man}(\text{Marcus})$
2. $\text{Pompeian}(\text{Marcus})$
3. $\neg \text{Pompeian}(x) \vee \text{Roman}(x)$
4. $\text{Ruler}(\text{Caesar})$
5. $\neg \text{Pompeian}(x) \vee \text{LoyalTo}(x, \text{Caesar}) \vee \text{Hates}(x, \text{Caesar})$
6. $\text{LoyalTo}(x, f(x))$ (Skolem function)
7. $\neg \text{Person}(x) \vee \neg \text{Ruler}(y) \vee \neg \text{TryAssassinate}(x, y) \vee \neg \text{LoyalTo}(x, y)$
8. $\text{TryAssassinate}(\text{Marcus}, \text{Caesar})$
9. $\text{Person}(x) \leftarrow (\text{need to add: } \forall x (\text{Man}(x) \rightarrow \text{Person}(x)) \rightarrow \neg \text{Man}(x) \vee \text{Person}(x))$
From 1 and $\neg \text{Man}(x) \vee \text{Person}(x)$: $\text{Person}(\text{Marcus})$

To prove: $\text{LoyalTo}(\text{Marcus}, \text{Caesar})$ is false. Assume $\text{LoyalTo}(\text{Marcus}, \text{Caesar})$ and derive contradiction.

Add negated goal: $\text{LoyalTo}(\text{Marcus}, \text{Caesar})$ (assume true to test consistency)

From 5 with $x = \text{Marcus}$: $\neg \text{Pompeian}(\text{Marcus}) \vee \text{LoyalTo}(\text{Marcus}, \text{Caesar}) \vee \text{Hates}(\text{Marcus}, \text{Caesar})$

From 2: $\text{LoyalTo}(\text{Marcus}, \text{Caesar}) \vee \text{Hates}(\text{Marcus}, \text{Caesar})$

With assumption $\text{LoyalTo}(\text{Marcus}, \text{Caesar})$, this is satisfied.

From 7 with $x = \text{Marcus}, y = \text{Caesar}$: $\neg \text{Person}(\text{Marcus}) \vee \neg \text{Ruler}(\text{Caesar}) \vee \neg \text{TryAssassinate}(\text{Marcus}, \text{Caesar}) \vee \neg \text{LoyalTo}(\text{Marcus}, \text{Caesar})$

From 9: $\text{Person}(\text{Marcus})$

From 4: $\text{Ruler}(\text{Caesar})$

From 8: $\text{TryAssassinate}(\text{Marcus}, \text{Caesar})$

This gives: $\neg \text{LoyalTo}(\text{Marcus}, \text{Caesar})$

This contradicts our assumption $\text{LoyalTo}(\text{Marcus}, \text{Caesar})$.

Therefore, **Marcus was NOT loyal to Caesar.**

What are the rules of conversion from predicate to CNF? Explain each rule with proper example. (10 marks)

Step-by-step conversion to CNF (Conjunctive Normal Form):

Rule 1: Eliminate implications

Replace $A \rightarrow B$ with $\neg A \vee B$

Example: $\forall x (\text{Man}(x) \rightarrow \text{Mortal}(x))$ becomes $\forall x (\neg \text{Man}(x) \vee \text{Mortal}(x))$

Rule 2: Move negations inward

Apply De Morgan's laws and eliminate double negations

- $\neg(A \wedge B) \rightarrow \neg A \vee \neg B$
- $\neg(A \vee B) \rightarrow \neg A \wedge \neg B$
- $\neg \forall x P(x) \rightarrow \exists x \neg P(x)$
- $\neg \exists x P(x) \rightarrow \forall x \neg P(x)$

Example: $\neg \forall x (P(x) \wedge Q(x)) \rightarrow \exists x (\neg P(x) \vee \neg Q(x))$

Rule 3: Standardize variables

Rename variables so each quantifier has a unique variable name

Example: $\forall x P(x) \vee \forall x Q(x)$ becomes $\forall x P(x) \vee \forall y Q(y)$

Rule 4: Skolemize

Remove existential quantifiers by replacing with Skolem constants/functions

Example: $\forall x \exists y \text{ Loves}(x, y)$ becomes $\forall x \text{ Loves}(x, f(x))$

Rule 5: Remove universal quantifiers

Drop all universal quantifiers (assume all variables are universally quantified)

Example: $\forall x (\neg \text{Man}(x) \vee \text{Mortal}(x))$ becomes $\neg \text{Man}(x) \vee \text{Mortal}(x)$

Rule 6: Distribute $\wedge$ over $\vee$

Convert to conjunctive normal form using: $A \vee (B \wedge C) \rightarrow (A \vee B) \wedge (A \vee C)$

Example: $(P \wedge Q) \vee R$ becomes $(P \vee R) \wedge (Q \vee R)$

Rule 7: Flatten nested conjunctions/disjunctions

Combine multiple $\wedge$ or $\vee$ operators

Example: $(A \vee B) \vee C$ becomes $A \vee B \vee C$

Write Short Note on: CNF (5 marks)

CNF (Conjunctive Normal Form) is a standardized representation of logical formulas where:

- The formula is a conjunction ($\wedge$) of clauses
- Each clause is a disjunction ($\vee$) of literals
- Each literal is an atomic proposition or its negation

Structure: $(L_{11} \vee L_{12} \vee \dots) \wedge (L_{21} \vee L_{22} \vee \dots) \wedge \dots$

Properties:

- Every propositional logic formula can be converted to CNF
- Used extensively in resolution theorem proving
- Simplifies automated reasoning algorithms

Example: $(\neg P \vee Q) \wedge (R \vee \neg S) \wedge (P \vee S \vee \neg Q)$

Conversion steps: Eliminate $\rightarrow$, move $\neg$ inward, distribute $\vee$ over $\wedge$.

Write Short Note on: Skolemization (5 marks)

Skolemization is the process of eliminating existential quantifiers from a first-order logic formula while preserving satisfiability.

Method:

- Replace each existentially quantified variable with a new function symbol (Skolem function) whose arguments are all universally quantified variables that scope over the existential quantifier
- For existential quantifiers not under any universal quantifier, use a Skolem constant

Purpose:

- Converts formulas to Skolem normal form (only universal quantifiers)
- Prepares formulas for resolution theorem proving

Example:

- $\exists x P(x) \rightarrow P(c)$ (Skolem constant)
- $\forall y \exists x P(x, y) \rightarrow \forall y P(f(y), y)$ (Skolem function)

Important: Skolemization preserves satisfiability but not logical equivalence.

Write Short Note on: Rules in Propositional Logic (5 marks)

Propositional Logic Rules for Inference:

Rule	Form	Description
Modus Ponens	$P, P \rightarrow Q \vdash Q$	If P is true and P implies Q, Q is true
Modus Tollens	$\neg Q, P \rightarrow Q \vdash \neg P$	If Q is false and P implies Q, P is false
Hypothetical Syllogism	$P \rightarrow Q, Q \rightarrow R \vdash P \rightarrow R$	Implication is transitive
Disjunctive Syllogism	$P \vee Q, \neg P \vdash Q$	If one disjunct is false, the other must be true
Conjunction	$P, Q \vdash P \wedge Q$	Two true statements can be conjoined
Simplification	$P \wedge Q \vdash P$	From a conjunction, derive either part
Addition	$P \vdash P \vee Q$	From a statement, derive its disjunction with anything
Resolution	$P \vee Q, \neg P \vee R \vdash Q \vee R$	Combine clauses eliminating complementary literals

Describe the process of knowledge representation using First Order Logic. (5 marks)

Process of Knowledge Representation in FOL:

Step 1: Identify objects and relations

- Define constants for specific objects (John, Book1)
- Define predicates for properties and relations (Loves, Red, FatherOf)

Step 2: Represent facts

Express basic knowledge as atomic sentences

Example: Loves(John, Mary) , Book(TaleOfTwoCities)

Step 3: Represent general rules

Use universal quantifiers ($\forall$) and implications

Example: $\forall x$ (Student(x) $\rightarrow$ AttendsClass(x))

Step 4: Represent existential knowledge

Use existential quantifiers ($\exists$) for existence claims

Example: $\exists x$ (Teacher(x) $\wedge$ Teaches(x, AI))

Step 5: Combine into knowledge base

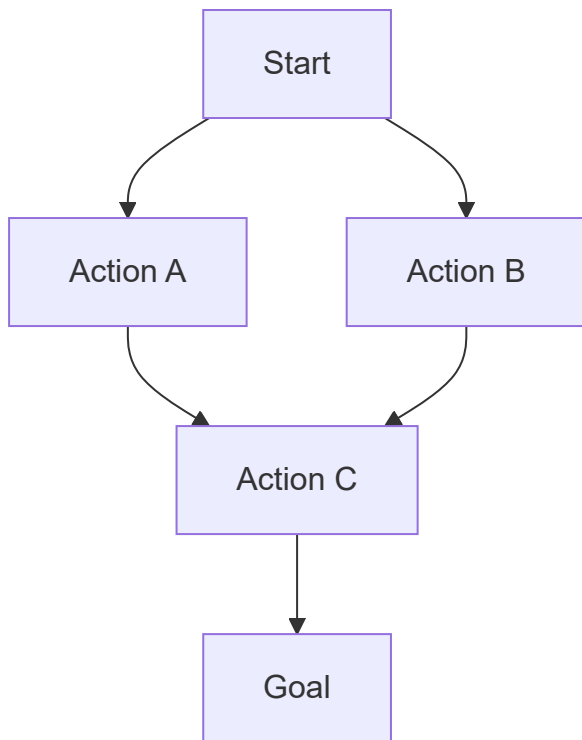
- Ensure consistent naming
- Avoid contradictions
- Test with query answering

Output: A formal, machine-interpretable set of sentences describing a domain.

Discuss Partial Order, Hierarchical, and Conditional Planning. (10 marks)

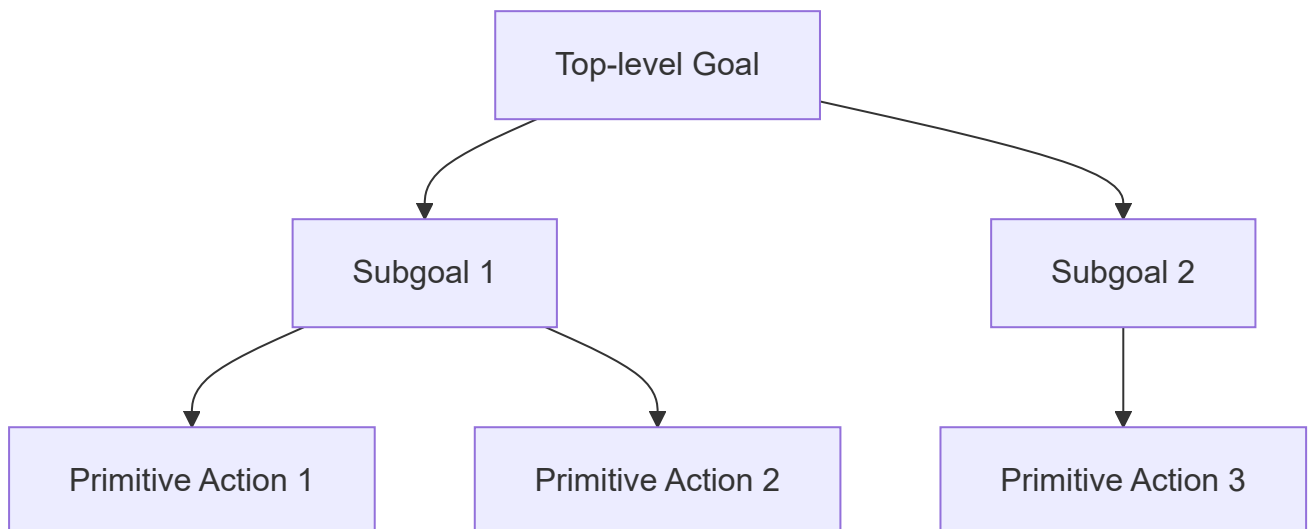
Partial Order Planning (POP):

- Plans represented as a set of actions with ordering constraints
- Actions can be executed in any order that satisfies constraints
- Uses "least commitment" principle - postpones decisions
- Maintains causal links between actions



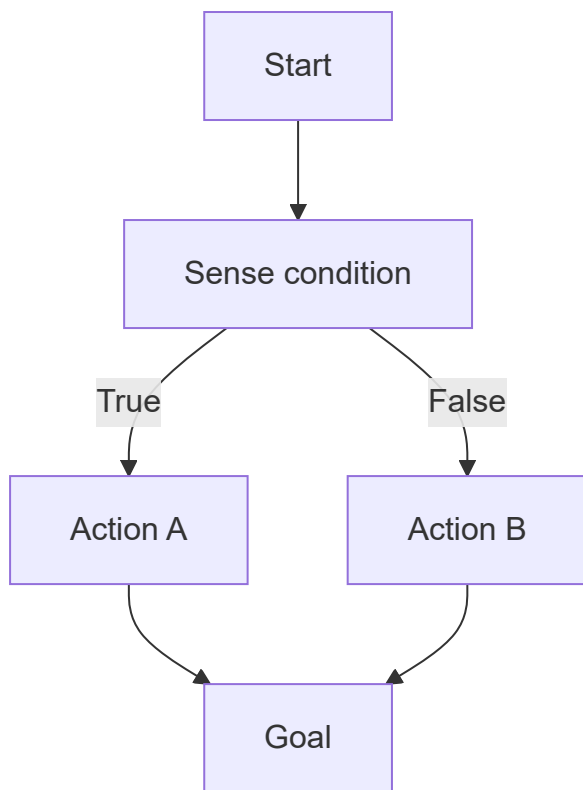
Hierarchical Planning:

- Plans decomposed into abstract levels
- Uses abstraction hierarchies to manage complexity
- Refines high-level actions into primitive actions



Conditional Planning:

- Handles uncertainty and partial observability
- Plans include conditional branches based on sensed information
- Uses "if-then-else" structures within plans



Comparison:

Aspect	Partial Order	Hierarchical	Conditional
Ordering	Flexible	Top-down	Branch-dependent
Complexity	Low to medium	Medium	High
Uncertainty	No	No	Yes
Main use	Scheduling	Complex tasks	Real-world planning

What is the difference between unification and Skolemization? (5 marks)

Aspect	Unification	Skolemization
Purpose	Find substitutions making expressions identical	Eliminate existential quantifiers
Operation	Matches variables with terms	Replaces variables with constants/functions
Preserves	Logical equivalence	Only satisfiability
Direction	Works on two expressions	Works on entire formula
Variables	Variables remain variables	Variables are replaced
Output	Most General Unifier (MGU)	Skolem normal form

Aspect	Unification	Skolemization
Use in resolution	Matches complementary literals	Preprocessing step before CNF

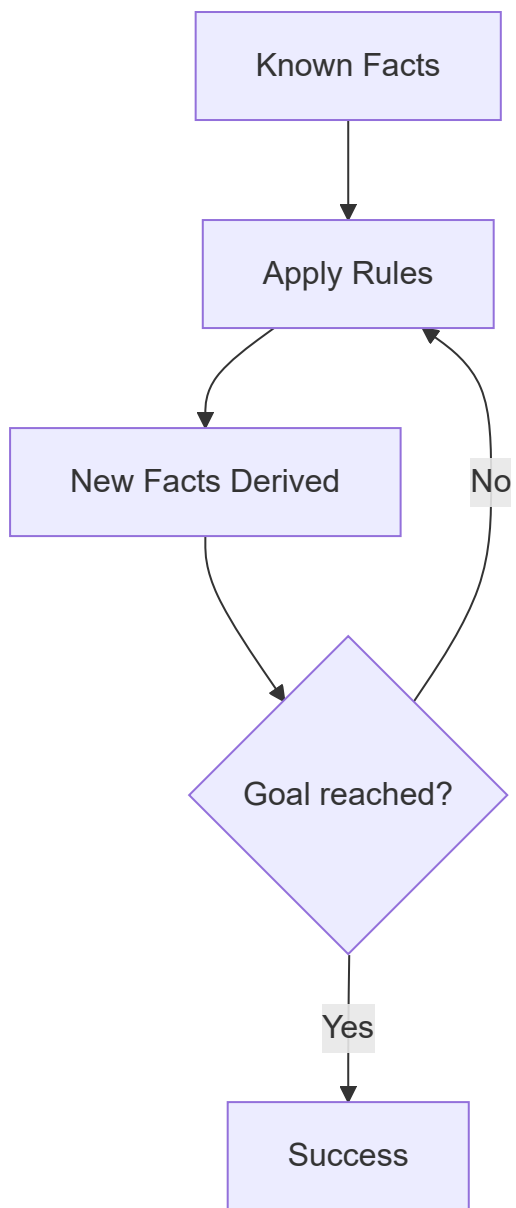
Example - Unification: $P(f(x), y)$ and $P(f(a), b)$ unify with $\{x/a, y/b\}$

Example - Skolemization: $\forall x \exists y Q(x, y) \rightarrow \forall x Q(x, f(x))$

Explain working of Forward and Backward Chaining algorithms. (5 marks)

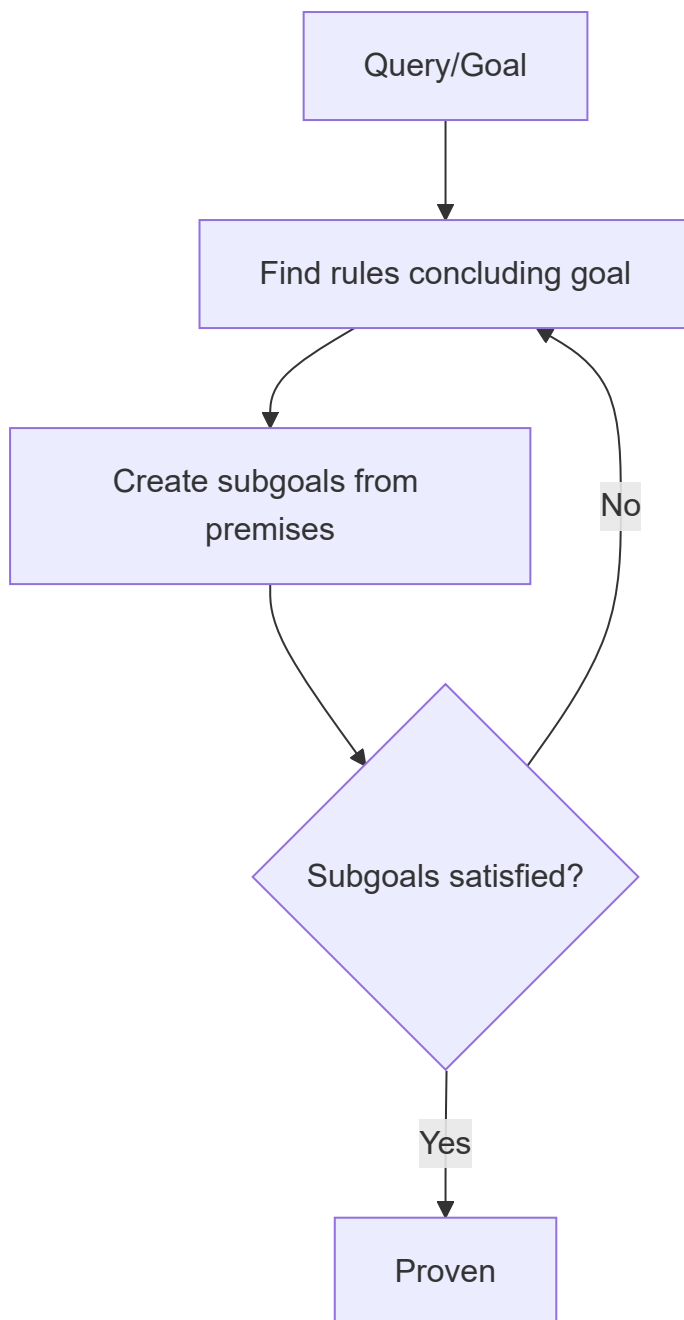
Forward Chaining (Data-driven):

- Starts from known facts
- Applies inference rules forward
- Continues until goal is reached or no more rules apply



Backward Chaining (Goal-driven):

- Starts from the goal
- Works backwards to find supporting facts
- Uses recursion/subgoaling



Comparison:

Aspect	Forward Chaining	Backward Chaining
Direction	Facts $\rightarrow$ Goal	Goal $\rightarrow$ Facts
Best for	Planning, monitoring	Diagnosis, query answering
Efficiency	May generate irrelevant facts	Focused on goal

Write Short Note on: Forward chaining based proofs (5 marks)

Forward Chaining Proofs are a direct method of theorem proving that starts with known axioms and facts, then repeatedly applies inference rules to derive new facts until the desired conclusion is proved.

Algorithm:

1. Initialize working memory with all known facts
2. Apply Modus Ponens using available rules
3. Add newly derived facts to working memory
4. Repeat until goal is derived or no new facts can be added

Properties:

- Sound (derives only true conclusions)
- Complete for definite clauses (Horn clauses)
- Data-driven - may derive many irrelevant facts

Example Knowledge Base:

- Facts: $\text{Dog}(\text{Rover})$, $\forall x (\text{Dog}(x) \rightarrow \text{Animal}(x))$
- Forward chaining: From $\text{Dog}(\text{Rover})$ and $\text{Dog}(x) \rightarrow \text{Animal}(x)$, derive $\text{Animal}(\text{Rover})$

Applications: Expert systems (OPS5, CLIPS), production rule systems, semantic reasoning.